Replay guarantee: minimal changes

Document 2 | Correct inequality; narrow the scope

Exact source text is quoted verbatim (LaTeX where applicable). Page references identify the saved before.pdf snapshot, not a newly compiled manuscript. All edits below are NOT APPLIED.

The wrong-direction prose remains in Related Works; the Method inequality is already correct. The proposals below correct the prose and clarify the local guarantee without changing the algorithm.

1. Correct the direction and its local scope — NOT APPLIED

Source: sections/related_works.tex:14–14 | before.pdf p. 2

OLD — exact source fragment:

RouterBSI ensures non-decreasing empirical error on the retained set while bounding replay evaluation costs.

NEW — exact replacement:

Including the unchanged predicate ensures non-increasing average error on the current replay set at each MBR selection, while bounding replay evaluation costs.

Why: Wrong direction: error must be non-increasing, not non-decreasing. Qualify the statement as local average risk on the same current set, not a temporal or per-example guarantee.

Grounding: sections/appendix/procedure_mbr.tex:16 (snapshot p. 12)

Current inequality — CORRECT / NO CHANGE

Source: sections/method.tex:60 | before.pdf p. 4, printed line 255

\[R_{\mathrm{replay}}(m_p^*\mid s_{fk})\leq R_{\mathrm{replay}}(p_f\mid s_{fk})\]

Meaning: selected average replay error is no greater than the unchanged predicate's error on the same replay set. This is not a guarantee for every task or future test performance.

Scope of the replay guarantee

2. Limit the problem objective — NOT APPLIED

Source: sections/problem_definition.tex:8–8 | before.pdf p. 3

OLD — exact source fragment:

to correct the observed error without degrading previously learned or unseen routing decisions.

NEW — exact replacement:

to reduce average replay error while using held-out validation to limit regressions.

Why: The minimum-risk choice need not correct the current error and may trade off previously correct decisions. Validation is relative to initialization, not a guarantee for unseen decisions.

Grounding: sections/appendix/procedure_mbr.tex:16 (snapshot p. 12); sections/appendix/procedure_validation.tex:19 (snapshot p. 12)

3. State the inequality's scope — NOT APPLIED

Source: sections/method.tex:60–60 | before.pdf p. 4

Existing anchor — retain unchanged (exact source):

\[R_{\mathrm{replay}}(m_p^*\mid s_{fk})\leq R_{\mathrm{replay}}(p_f\mid s_{fk})\]

INSERT immediately after anchor:

This inequality concerns average error on the current replay set; it does not guarantee preservation of every previously correct decision or improvement on unseen tasks.

Why: Keep the existing inequality unchanged; append this one sentence to expose its already-correct, narrow scope.

Grounding: sections/appendix/procedure_mbr.tex:16 (snapshot p. 12); sections/appendix/procedure_validation.tex:19 (snapshot p. 12)

Scope of the validation gate

4. Name the fixed comparison baseline — NOT APPLIED

Source: sections/method.tex:80–80 | before.pdf p. 4

Existing anchor — retain unchanged (exact source):

\[\begin{aligned}&g(T_v,m_p^*,p_{si})\\&\quad=\mathds{1}[R_v(m_p^*\mid s_{fk})< R_v(p_{si}\mid s_{fk})]\ end{aligned}\]

INSERT immediately after anchor:

The gate compares with the fixed cold-start predicate, not the previous accepted predicate, so validation error need not improve monotonically across iterations.

Why: A single scope sentence prevents reading the cold-start gate as a guarantee of improvement over the last policy.

Grounding: sections/appendix/procedure_validation.tex:19 (snapshot p. 12)

5. Align Figure 2 with strict admission (optional) — NOT APPLIED

Source: sections/related_works.tex:8–8 | before.pdf p. 3

OLD — exact source fragment:

Selected candidate is discarded by the held-out validation gate if its validation error is higher than the initial predicate $\ mathrm{p}_{\mathrm{si}}$.

NEW — exact replacement:

Selected candidate is discarded by the held-out validation gate if its validation error is higher than or equal to that of the initial predicate p_{si}.

Why: Compact caption alignment with the strict admission rule already stated in the method; equality must also be rejected.

Grounding: sections/appendix/procedure_validation.tex:12 (snapshot p. 12)

No change is proposed to the minimum-risk equation, the validation-gate equation, or the author-confirmed shared initialization.